$$g(w)=\cos(3w)sec(1-w)$$
$$g'(w)=-\sin(3w).3\,sec(1-w)+\cos(3w).\,sec(1-w)\tan(1-w).-1$$
$$g'(w)=-(3\sin(3w)sec(1-w)+\cos(3w)sec(1-w)\tan(1-w))$$